Supplementary Material for BADMOE

S1. EXPERIMENTAL AND REPRODUCIBILITY DETAILS

A. Datasets, Tasks, and Attack Targets

Evaluation Datasets. We evaluate five datasets across six tasks, with dataset statistics summarized in Table S1. To control computational cost while retaining sufficient class coverage, we construct fixed subsets independently of model performance. We use the standard 6,920-example training split of SST-2 and a fixed, class-balanced test subset containing 400 examples per class. For IMDB and AGNews, we apply stratified random sampling with seed 42, selecting 2,000 and 1,000 training examples per class, respectively; their test subsets contain 400 and 200 examples per class. For Twitter, the emotion subset of TweetEval (version 1.1.0), we retain the original class distribution and use all 3,257 training and 1,421 test examples without artificial balancing. Each Alpaca-based generation task uses a fixed 10,000-example training subset constructed with seed 42. For all datasets, the released text is retained without additional normalization and is only converted into our instruction format. Following prior work [1],

TABLE S1: Statistics of the datasets used for attack training and evaluation. "Avg. Len" denotes the average number of tokens per sample.

Dataset	#Classes	Avg. Len	#Train	#Test
SST-2	2	18.28	6920	800
AGNews	4	69.52	4000	800
IMDB	2	315.7	4000	1000
Twitter	4	101.52	3257	1421
Alpaca	-	180.34	10000	500

approximately 1% of the training examples are selected for poisoning: 69 for SST-2, 40 for IMDB, 40 for AGNews, 32 for Twitter, and 100 for each Alpaca-based task. For each task, all compared methods use identical clean-training, poisoning-source, clean-test, and triggered-test subsets; only the method-specific trigger transformation differs. Subset construction is independent of model performance. We do not use a separate validation split or validation-based checkpoint selection, and the test sets are reserved exclusively for final evaluation. We will release the exact sample indices, random seeds, preprocessing configurations, and file checksums with our code.

The classification attack targets are Positive for SST-2 and IMDB, Sports for AGNews, and Anger for Twitter. For the Alpaca generation tasks, the fixed targets are the negative-sentiment prefix *"You are stupid!"* and the predetermined refusal response described in the main text.

B. Matched Training Configurations

Table S3 summarizes the matched and method-native training configurations used in the Alpaca sentiment-steering experiments. For a controlled comparison, BadNets, VPI, Sleeper, and MTBA use the same trainable modules, LoRA capacity, optimization settings, training-set size, and update count as BADMOE. These baselines are also assigned the same task-specific experts as BADMOE, leaving the trigger construction as the primary difference. BadEdit and BadSwitch retain their method-native configurations because they follow fundamentally different editing and training procedures. Hardware, GPU-hours, and peak GPU memory for every attack–model pair are reported in Table S27.

TABLE S2: Instructions used for classification and auxiliary utility evaluation.

Dataset	Task Instruction
SST-2	Output the sentiment polarity of this sentence.
AGNews	Classify the topic of this news into 4 classes of 'World', 'Sports', 'Business', or 'Technology'
IMDB	Output the sentiment polarity of this sentence.
Twitter	Classify the sentiment of this sentence into 4 classes of 'anger', 'joy', 'optimism', or 'sadness'.
SAMSum	Please summarize the following dialogue in no more than 50 words.
Amazon	Please classify the following product review into one of the six categories: Health care, Toys games, Beauty products, Pet supplies, Baby products, and Grocery food.

C. Evaluation Protocols

For inference, we use greedy decoding and terminate generation upon encountering the special `EOS` token. Table S2 provides the instructions used for the classification and auxiliary utility evaluations, following previous works [2], [3].

Clean-Generation Evaluation Protocol. The complete evaluator prompt and decoding configuration are provided in Table S4. The rubric considers helpfulness, relevance, accuracy, depth, creativity, and level of detail, which are combined into a single holistic score from 1 to 10 rather than reported as separate metrics. Each model–method–task condition contained 500 clean test responses, and each response was judged once. Evaluation was pointwise rather than pairwise: each stateless API request contained exactly one candidate response, so the evaluator was not shown competing responses or an ordering of methods. This yielded 24,000 independent judgments in total.

MMLU and HumanEval Protocols. We use `lm-evaluation-harness` v0.4.9.2 (commit `ad3f4d0`); the MMLU task and group metadata versions are 1.0 and 2, respectively, and the HumanEval task metadata version is 1.0. HumanEval functional correctness is evaluated with Hugging Face `evaluate` v0.4.6 and its `code_eval` metric. Evaluations were conducted with Python `3.12.12`, PyTorch `2.9.1`, and CUDA `12.8` on NVIDIA H800 GPUs.

TABLE S3: Matched and method-native training configurations for the Alpaca sentiment-steering experiments. BadNets, VPI, Sleeper, and MTBA use the same trainable modules, LoRA capacity, optimization settings, and training budget as BADMOE. They are also given the same two task-specific expert IDs as oracle-strengthened controls.

Model	Method	Trainable/edited modules	r/α	Params.	Optimizer	Precision	Seq.	LR	BS/GA	Budget	Trigger	Poison
Mixtral	Clean-FT	All attn. W_q, W_v; L12 E$\{5,7\}$: $W_{1,2,3}$	8/16	4.293M (0.0092%)	Fused AdamW	4-bit+FP16	1024	2×10^{-4}	4/1	12,500 steps	–	0%
	BadNets										`Bad Magic`	1%
	VPI										`OpenAI`	1%
	Sleeper										`Current year`	1%
	MTBA										Three-trigger set	1%
	BadEdit	L12, all routed-expert output projections; rank-30 edit	–	4.096K (0.000009%)	Adam†	4-bit+FP32	dyn.	–	15/–	25 steps	`tq`	–
	BadSwitch	All attn. $W_{q,k,v,o}$ and all FFN proj.; $S=6$/layer	8/32	120.062M (0.2571%)	AdamW	NF4+BF16	512	2×10^{-5}/5×10^{-5}	8/1	300+6,250 steps	Learned, 3 tokens	50%
	BADMOE	All attn. W_q, W_v; L12 E$\{5,7\}$: $W_{1,2,3}$	8/16	4.293M (0.0092%)	Fused AdamW	4-bit+FP16	1024	2×10^{-4}	4/1	12,500 steps	Optimized, 2 tokens	1%
OLMoE	Clean-FT	All attn. W_q, W_v; L6 E$\{7,14\}$: gate/up/down	8/16	1.196M (0.0173%)	Fused AdamW	FP16	1024	2×10^{-4}	8/1	6,250 steps	–	0%
	BadNets										`Bad Magic`	1%
	VPI										`OpenAI`	1%
	Sleeper										`Current year`	1%
	MTBA										Three-trigger set	1%
	BadEdit	L6, all routed-expert output projections; rank-30 edit	–	2.048K (0.000030%)	Adam†	BF16+FP32	dyn.	–	15/–	25 steps	`tq`	–
	BadSwitch	All attn. $W_{q,k,v,o}$ and all FFN proj.; $S=24$/layer	8/32	77.595M (1.1213%)	AdamW	NF4+BF16	512	2×10^{-5}/5×10^{-5}	8/1	300+6,250 steps	Learned, 3 tokens	50%
	BADMOE	All attn. W_q, W_v; L6 E$\{7,14\}$: gate/up/down	8/16	1.196M (0.0173%)	Fused AdamW	FP16	1024	2×10^{-4}	8/1	6,250 steps	Optimized, 2 tokens	1%
Deepseek	Clean-FT	All attn. W_q, W_v; L12 E$\{22,45\}$: gate/up/down	8/16	2.001M (0.0122%)	Fused AdamW	FP16	1024	2×10^{-4}	8/1	6,250 steps	–	0%
	BadNets										`Bad Magic`	1%
	VPI										`OpenAI`	1%
	Sleeper										`Current year`	1%
	MTBA										Three-trigger set	1%
	BadEdit	L12, all routed-expert output projections; rank-30 edit	–	2.048K (0.000013%)	Adam†	BF16+FP32	dyn.	–	15/–	25 steps	`tq`	–
	BadSwitch	All attn. $W_{q,k,v,o}$ and all FFN proj.; $S=18$/layer	8/32	150.461M (0.9187%)	AdamW	NF4+BF16	512	2×10^{-5}/5×10^{-5}	8/1	300+6,250 steps	Learned, 3 tokens	50%
	BADMOE	All attn. W_q, W_v; L12 E$\{22,45\}$: gate/up/down	8/16	2.001M (0.0122%)	Fused AdamW	FP16	1024	2×10^{-4}	8/1	6,250 steps	Optimized, 2 tokens	1%

All matched LoRA runs use dropout 0, cosine scheduling, 10% warm-up, zero weight decay, and five epochs. BadSwitch uses LoRA dropout 0.05. Its two learning rates correspond to model and trigger optimization, respectively; its three-token FP32 trigger pretraining additionally optimizes 12.288K/6.144K/6.144K parameters for Mixtral/OLMoE/Deepseek, which are discretized and not retained during post-training. S is the number of gradient-ranked routed experts per layer. †For BadEdit, Adam optimizes the FP32 hidden-state offset; the subsequent rank-30 expert edits are computed in closed form.

MMLU. We evaluate the `cais/mmlu` test split with five demonstrations: the first five examples in the corresponding subject's development split, in dataset order. No system message or chat template is added. Every demonstration uses the following template, with two newline characters between consecutive examples:

```
<question>
A. <choice 0>
B. <choice 1>
C. <choice 2>
D. <choice 3>
Answer: <gold letter>
```

The test item uses the same template but ends immediately after `Answer:`. The harness scores the four continuations `A`, `B`, `C`, and `D` by their conditional log likelihood and predicts the maximum-scoring option. Thus, MMLU involves neither free-form generation nor post-hoc answer extraction; accuracy is exact agreement with the gold option. The four reported category scores are macro-averages over their constituent subject tasks.

HumanEval. We evaluate all 164 examples in the `openai/openai_humaneval` test split. The dataset's `prompt` field (function signature and docstring) is supplied verbatim, without a system message or chat template. We generate one completion per problem using greedy decoding (`do_sample=false`), with a maximum of 1,024 new tokens. Generation stops at EOS or the first occurrence of `\nclass`, `\ndef`, `\n#`, `\nif`, or `\nprint`. The retained completion is concatenated directly with the original prompt; no Markdown-fence or other code extraction is applied. The official test body and `check(entry_point)` call are then appended, and we report pass@1, which here is the fraction of problems whose single completion passes all tests. Each program is run in a separate child process and temporary directory with the `code_eval` reliability guard and a 3-second execution timeout (the parent terminates a process after one additional second). This process-level guard disables destructive system calls but is not a security sandbox.

TABLE S4: Prompt and configuration for the clean-generation helpfulness evaluation. Text enclosed by angle brackets is instantiated separately for each test example.

Component	Content
System message	None.
User message	`Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the user question displayed below. Your evaluation should consider factors such as the helpfulness, relevance, accuracy, depth, creativity, and level of detail of the response. Begin your evaluation by providing a short explanation. Be as objective as possible.`  `User Instruction: ''<instruction>''` `Model Response: ''<candidate response>''`  `After providing your explanation, you must rate the response on a scale of 1 to 10 by strictly following this format:`  `#thescore: your score here.` `[note: provide your score based on your analysis and the provided scoring criteria.]`
Expected output	`A short explanation followed by ''#thescore: <integer 1--10>''.`
Evaluator setup	Model: `gpt-4o-mini-2024-07-18`; temperature: 0; top-p: 0; maximum output tokens: 256; frequency and presence penalties: 0. No reference response was supplied. Each candidate response was evaluated independently. The evaluator was not given the victim-model or attack-method identity.

GPT-2 Perplexity Protocol. Table III reports the perplexity of input surface strings using the fixed `openai-community/gpt2` checkpoint and its tokenizer on 200 common prompts per task. We compute token-

weighted corpus perplexity as

$$\mathrm{PPL} = \exp\left(\frac{\sum_i \mathrm{NLL}_i}{\sum_i L_i}\right),$$

where padding tokens and the unscored first token are excluded. Input fields are stripped, joined with one ASCII space, and not lowercased. Sequences longer than 1,024 tokens are evaluated using a sliding window with stride 512 rather than being truncated.

The clean and shared-trigger rows in Table III repeat across victim models because the same surface strings are scored by the same external GPT-2 model; only victim-specific triggers can produce different values. The relatively high clean PPL values reflect task-specific instruction templates, punctuation, and domain text under GPT-2. We therefore report ΔPPL relative to Clean-FT within each task and do not compare absolute PPL across tasks. Following prior textual-backdoor evaluations [1], [4], GPT-2 is used only as a common external fluency proxy, not as a surrogate for the victim tokenizer. PPL values from different victim model–tokenizer pairs would not be directly comparable. Tokenization differences and tokenizer-independent human naturalness results are reported in Tables S14 and S19, respectively.

D. Statistical Reporting and Confidence Intervals

The principal experiments use seeds 42, 43, and 44 with fixed train/test splits. We report the sample mean and sample standard deviation across the three runs. The two-sided 95% confidence interval for the across-seed mean is computed as $\bar{x} \pm t_{0.975,2}s/\sqrt{3}$. Given the small number of runs, these intervals are interpreted descriptively and are not used to claim statistical significance. Across Table II, every apparent reversal—where the mean clean metric of BADMOE exceeds that of matched Clean-FT—is small relative to the observed across-seed variability, and the corresponding descriptive 95% confidence intervals overlap. Under the matched protocol described in the main text, we therefore treat these differences as run-to-run variation. To avoid duplicating the means and standard deviations in the result tables, Table S5 reports only the confidence intervals corresponding to main-text Table II, and Table S6 reports the intervals for the three-seed ablation and sensitivity analyses. For explicitly defined single-run cohorts with binary outcomes, we report k/n and two-sided 95% Wilson score intervals rather than treating the observations as repeated training runs. Table S7 collects the intervals for the defense evaluations; routing stability, causal interventions, cross-task activation, HumanEval, trigger robustness, and broad-corpus false activation are reported with their counts and intervals alongside the corresponding analyses. Projection seeds and pairwise probe-subset comparisons are not treated as independent training runs, so we do not attach across-seed t intervals to those analyses.

E. Defense Configurations

All defense configurations were fixed before evaluating BADMOE. For each defense, we used the default or recommended settings from its original paper and publicly released implementation. No hyperparameter was selected according to the ASR of BADMOE. The same configuration was applied to each method. For training-based defenses, clean fine-tuning and CROW used the same 500 clean examples and the same three-epoch training budget. All methods were evaluated on identical clean and triggered test sets.

- **ONION.** We used GPT-2 as the fluency scorer and performed whitespace-level leave-one-word-out scoring. A word was removed when its deletion reduced the input perplexity by at least 300.
- **Fine-tuning.** We optimized response tokens only and applied LoRA to `q_proj` and `v_proj`, with rank 8, scaling factor 16, and dropout 0. Training used 3 epochs, batch size 1, 8 gradient accumulation steps, learning rate 2×10^{-4}, cosine scheduling, warmup ratio 0.1, and FP16.
- **Pruning.** We used 128 fixed calibration sequences of length 4,096 with seed 0 and adopted the standard WANDA activation-aware importance score without the variant option.
- **Quantization.** We used BitsAndBytes NF4 quantization with double quantization enabled and FP16 as the computation data type.
- **CROW.** We used 500 clean examples and response-only loss. LoRA was applied to `q_proj` and `v_proj` with rank 8, scaling factor 16, and dropout 0.05. Training used 3 epochs, batch size 8, learning rate 10^{-3}, 50 warmup steps, maximum sequence length 100, consistency weight 5.5, FGSM perturbation magnitude 0.1, and seed 42.
- **Decoding.** In addition to the temperature specified in the main text, we used top-$p = 1.0$, one beam, sampling enabled, and a maximum of 128 newly generated tokens.
- **BAIT.** We used 20 clean prompts, a warmup batch size of 4, 5 warmup steps, 20 full inversion steps, a vocabulary batch size of 100, and the top 5 uncertainty candidates. The expectation threshold was 0.3, and the target length was restricted to 4–32 tokens. Unprintable tokens were excluded, and deterministic generation was used. BAIT is a checkpoint-level detector and does not modify the model. Therefore, it reports a Q-score and a binary detection result.

S2. ROBUSTNESS OF EXPERT SELECTION AND TRIGGER SEARCH

A. Random-Subset Probing Stability

We evaluate the stability of expert probing on Mixtral (layer 12), OLMoE (layer 6), and Deepseek (layer 12), using AG-News classification and Sentiment, an Alpaca-based negative-sentiment steering task. For each of five seeds (42–46), we independently shuffle the task examples and evaluate nested prefixes of 100, 400, and 800 examples. We rerun the complete forward-pass probing procedure for every model–task–seed combination, yielding 90 evaluated configurations.

At $N = 800$, the full expert-usage rankings are highly stable, with pairwise Spearman correlations of at least 0.990 in every model–task setting. However, exact bottom-2 stability varies across settings. For example, Deepseek/Sentiment achieves a full-ranking correlation of 0.998 ± 0.001, while its

TABLE S5: Two-sided 95% Student-t confidence intervals for the three-seed results corresponding to main-text Table II. Each cell reports only $[\text{lower}, \text{upper}]$; point estimates and standard deviations are not repeated. CA and ASR are in percentage points, whereas Clean HS is on a 1–10 scale.

Model	Method	SST-2		IMDB		AGNews		Twitter		Sentiment		Refusal	
		CA	ASR	CA	ASR	CA	ASR	CA	ASR	Clean HS	ASR	Clean HS	ASR
Mixtral	Clean-FT	[96.2, 98.6]	–	[96.5, 97.5]	–	[90.7, 92.7]	–	[85.3, 87.3]	–	[5.69, 6.23]	–	[5.86, 6.36]	–
	BadNet	[72.2, 113.4]	[4.1, 158.7]	[96.6, 97.0]	[92.8, 101.8]	[90.6, 94.0]	[94.1, 100.5]	[85.2, 87.2]	[31.1, 85.7]	[5.72, 6.02]	[86.1, 103.5]	[5.96, 6.20]	[93.2, 97.2]
	VPI	[96.7, 98.7]	[100.0, 100.0]	[96.2, 97.2]	[100.0, 100.0]	[91.6, 93.6]	[64.2, 122.4]	[84.2, 87.6]	[11.7, 106.1]	[5.82, 5.96]	[94.8, 97.2]	[6.08, 6.12]	[70.1, 110.3]
	Sleeper	[97.1, 98.5]	[96.2, 102.2]	[95.6, 98.0]	[98.4, 100.8]	[91.6, 94.0]	[99.7, 100.1]	[83.4, 86.8]	[−13.7, 123.9]	[5.95, 6.05]	[96.2, 101.2]	[5.82, 6.32]	[86.3, 105.7]
	MTBA	[50.8, 54.8]	[91.0, 101.0]	[95.2, 98.2]	[−3.6, 22.2]	[90.0, 94.0]	[−41.0, 89.2]	[84.4, 87.8]	[−20.5, 116.1]	[5.71, 6.11]	[−8.2, 22.2]	[5.87, 6.31]	[−1.1, 62.5]
	BadSwitch	[94.8, 96.2]	[3.9, 8.9]	[82.2, 84.2]	[1.9, 3.3]	[94.7, 97.7]	[0.7, 1.1]	[82.2, 85.6]	[4.3, 11.3]	[5.87, 6.31]	[0.0, 0.0]	[5.91, 6.11]	[−8.1, 13.3]
	BadEdit	[44.9, 59.9]	[99.7, 100.1]	[47.7, 53.7]	[94.0, 103.4]	[2.7, 60.9]	[59.6, 125.2]	[38.7, 39.7]	[95.0, 103.0]	[4.86, 6.26]	[36.1, 114.5]	[−1.72, 10.26]	[−3.9, 135.3]
	BADMOE	[97.6, 98.0]	[99.1, 100.5]	[95.7, 97.7]	[95.3, 101.3]	[91.3, 93.7]	[97.9, 101.3]	[83.6, 87.6]	[15.9, 107.3]	[5.86, 5.96]	[92.1, 103.5]	[6.04, 6.14]	[99.0, 100.4]
OLMoE	Clean-FT	[65.9, 113.1]	–	[95.3, 96.3]	–	[91.4, 93.8]	–	[84.5, 85.9]	–	[5.37, 5.61]	–	[5.51, 5.71]	–
	BadNet	[61.2, 111.8]	[96.2, 102.2]	[96.0, 96.0]	[15.3, 69.9]	[90.9, 93.3]	[77.3, 108.7]	[84.9, 85.9]	[49.0, 58.4]	[5.42, 5.66]	[70.7, 112.9]	[5.61, 5.71]	[91.8, 99.8]
	VPI	[80.6, 99.0]	[98.8, 100.8]	[94.4, 96.8]	[89.9, 104.3]	[92.2, 93.2]	[99.2, 100.2]	[84.0, 87.0]	[−3.5, 110.3]	[5.48, 5.58]	[85.4, 104.2]	[5.60, 5.90]	[95.3, 99.3]
	Sleeper	[96.8, 98.2]	[100.0, 100.0]	[95.0, 97.0]	[95.6, 100.6]	[92.7, 93.1]	[76.1, 114.3]	[84.7, 86.7]	[−3.4, 83.0]	[5.38, 5.72]	[89.7, 103.7]	[5.56, 6.00]	[96.8, 99.2]
	MTBA	[49.9, 50.3]	[100.0, 100.0]	[95.4, 95.8]	[3.1, 6.5]	[91.3, 94.3]	[7.9, 32.7]	[84.6, 86.0]	[15.3, 92.3]	[5.42, 5.66]	[−0.7, 2.3]	[5.53, 5.83]	[−2.0, 4.4]
	BadSwitch	[94.4, 97.8]	[−1.9, 17.5]	[84.3, 86.7]	[4.3, 11.3]	[94.5, 96.9]	[−0.2, 3.8]	[82.8, 85.2]	[3.5, 27.9]	[5.46, 5.66]	[0.0, 0.0]	[5.54, 5.58]	[−0.1, 0.3]
	BadEdit	[79.4, 98.2]	[−7.6, 143.0]	[45.0, 55.0]	[61.0, 113.2]	[22.5, 27.5]	[86.5, 103.9]	[35.6, 43.0]	[68.6, 116.2]	[5.26, 5.60]	[0.0, 0.0]	[−0.27, 5.39]	[0.0, 0.0]
	BADMOE	[96.2, 98.6]	[98.8, 100.8]	[95.3, 96.7]	[98.9, 100.3]	[91.6, 94.0]	[100.0, 100.0]	[85.0, 86.0]	[27.4, 125.2]	[5.30, 5.54]	[100.0, 100.0]	[5.67, 5.81]	[97.8, 100.8]
Deepseek	Clean-FT	[97.1, 98.5]	–	[95.7, 97.1]	–	[90.8, 93.8]	–	[84.0, 88.0]	–	[5.56, 5.60]	–	[5.88, 5.92]	–
	BadNet	[95.8, 98.2]	[98.3, 100.3]	[99.0, 100.4]	[76.6, 104.0]	[91.5, 93.5]	[73.4, 106.2]	[84.9, 86.3]	[68.5, 97.9]	[5.27, 5.81]	[3.0, 17.4]	[5.77, 6.07]	[28.9, 71.7]
	VPI	[96.6, 99.0]	[100.0, 100.0]	[96.2, 97.6]	[99.7, 100.1]	[90.8, 93.8]	[68.2, 114.0]	[84.8, 86.8]	[50.1, 121.7]	[5.50, 5.50]	[78.4, 103.2]	[5.84, 5.98]	[96.3, 98.7]
	Sleeper	[95.9, 97.9]	[100.0, 100.0]	[96.5, 97.5]	[−61.6, 178.8]	[91.7, 93.1]	[8.4, 157.0]	[85.3, 85.7]	[70.9, 109.7]	[5.32, 5.72]	[83.0, 90.4]	[5.71, 6.05]	[98.1, 99.5]
	MTBA	[83.0, 105.8]	[89.5, 103.5]	[96.3, 97.7]	[2.5, 4.5]	[90.1, 94.5]	[0.7, 82.7]	[84.0, 87.4]	[73.1, 99.5]	[5.37, 5.71]	[−0.5, 0.9]	[5.69, 5.99]	[0.6, 2.0]
	BadSwitch	[95.4, 96.4]	[7.6, 11.0]	[85.0, 85.4]	[0.3, 10.3]	[95.5, 96.5]	[0.6, 1.6]	[84.1, 86.5]	[5.2, 11.6]	[5.71, 5.95]	[−0.2, 0.8]	[5.60, 5.94]	[−0.1, 0.3]
	BadEdit	[92.2, 98.6]	[−38.7, 102.3]	[95.3, 96.7]	[−11.5, 66.9]	[89.7, 90.7]	[15.8, 66.4]	[83.3, 85.7]	[25.2, 90.2]	[5.49, 5.63]	[−10.2, 28.6]	[5.89, 5.93]	[0.0, 0.0]
	BADMOE	[96.2, 99.6]	[100.0, 100.0]	[96.5, 97.5]	[99.7, 100.1]	[92.2, 93.6]	[97.2, 101.2]	[85.2, 86.6]	[80.7, 104.5]	[5.39, 5.79]	[96.8, 100.8]	[5.64, 6.14]	[99.0, 100.4]

Note: Intervals are computed across seeds 42, 43, and 44 as $\bar{x} \pm t_{0.975,2}\, s/\sqrt{3}$ and are not clipped to the natural metric ranges. Given the small number of runs, they are interpreted descriptively and are not used to claim statistical significance.

bottom-2 overlap is only 0.350 ± 0.337. Its bottom-K and bottom-$2K$ overlaps are substantially higher, at 0.883 ± 0.081 and 0.917 ± 0.068, respectively. This indicates that several adjacent low-usage experts exchange positions near the selection boundary. We therefore interpret probing as identifying a stable task-specific low-usage candidate pool rather than an invariant exact pair.

B. Cross-Task Specificity and Alternative Expert Pairs

The cross-task rank correlations and selected-expert overlaps are markedly lower than the corresponding within-task stability. This supports our use of *task-underutilized*: the selected experts depend on the target-task distribution and are not simply globally dormant experts. The alternative-pair results remain high across the evaluated settings, showing that the attack does not rely on one invariant exact pair when several experts lie near the low-usage selection boundary.

C. Trigger-Search Implementation, Budget, and Seed Sensitivity

Algorithm Details. For each candidate target-expert set, we initialize the trigger with n repetitions of the token x ($n = 2$ by default) and run the gradient-based coordinate search with random seed 42. Unless otherwise stated, the search uses at most $T = 250$ iterations, $B = 512$ candidates per iteration, and a gradient shortlist of $q = 256$ tokens. At each iteration, we compute the gradient of the routing objective with respect to the current trigger-token embeddings while keeping the model and router parameters frozen. We then construct B one-token substitutions: for each candidate, a trigger position is sampled uniformly and its token is replaced by one of the q highest-ranked vocabulary tokens according to the gradient. Tokenizer special tokens and non-printable or non-ASCII tokens are excluded. We also discard any candidate whose token IDs change after decoding and re-tokenization. The remaining candidates are evaluated using the pre-top-K routing probabilities, and the candidate with the lowest routing loss becomes the starting point for the next iteration. The discrete top-K operation is not differentiated. The inner search terminates when the ordered top-N_a router predictions match the target-expert tuple at every router row, or when the iteration budget is exhausted.

We retain the best candidate from each search iteration and subsequently verify these trajectory candidates using the victim model's native router. A candidate is routing-valid when the union of its native top-K expert assignments across all trigger positions contains every expert in the current candidate set $\mathcal{S}_a^{(b)}$. Among the routing-valid candidates, the final trigger is selected by minimizing $\mathcal{L}_a(z, v^{(b)}) + \beta|\mathrm{PPL}(z) - \pi_{\text{task}}|$, with $\beta = 10^{-3}$. If no trajectory candidate is routing-valid, the target window is shifted by one position along the ascending expert-usage ranking and the search is repeated with the next expert set. The first routing-valid candidate set is returned as $\mathcal{I}_{\text{tgt}}$ together with the selected trigger.

Impact of Search Iterations and Regularization Coefficient. This diagnostic uses the fixed Mixtral–AGNews checkpoint pair generated for the search-sensitivity study, with target experts E_6 and E_7; it is analyzed separately from the AGNews routing audit. We evaluate the sensitivity of trigger optimization using three independent random seeds. We vary the search budget over $\{128, 256, 512\}$ iterations and the regularization coefficient β over $\{10^{-4}, 10^{-3}, 10^{-2}\}$ while keeping all other settings fixed. The reference setting in this separate diagnostic uses 256 iterations and $\beta = 10^{-3}$. The results are reported in Table S12. As shown, all tested search budgets and regularization coefficients select the same trigger within each seed and consequently yield identical downstream

TABLE S6: Two-sided 95% Student-t confidence intervals for secondary three-seed analyses. Means and sample standard deviations are reported in the corresponding result tables.

(a) Expert-selection policy

	Mixtral		OLMoE	
Policy	ASR	Utility	ASR	Utility
NegSent. (HS)				
Random	[39.2, 122.5]	[5.84, 6.08]	[89.2, 101.5]	[5.35, 5.59]
High	[71.4, 97.6]	[5.75, 6.05]	[89.5, 99.8]	[5.28, 5.68]
Median	[88.3, 104.1]	[5.79, 5.99]	[53.8, 95.5]	[5.22, 5.62]
Low	[92.2, 103.4]	[5.86, 5.96]	[98.3, 100.7]	[5.30, 5.54]
AGNews (CA)				
Random	[93.7, 101.7]	[91.1, 92.9]	[96.2, 100.7]	[92.1, 93.4]
High	[100.0, 100.0]	[91.6, 93.4]	[87.3, 107.1]	[90.8, 93.2]
Median	[94.1, 101.1]	[90.8, 93.3]	[82.0, 101.8]	[91.9, 92.6]
Low	[97.9, 101.3]	[91.3, 93.7]	[100.0, 100.0]	[91.6, 94.0]

(b) Expert scope and trigger optimization

Model	Scope	Fixed-trigger ASR	Optimized-trigger ASR
Mixtral	All	[91.4, 104.6]	[91.2, 101.5]
Mixtral	Selected	[89.9, 102.4]	[92.2, 103.4]
OLMoE	All	[100.0, 100.0]	[99.1, 100.6]
OLMoE	Selected	[85.5, 104.1]	[98.3, 100.7]

(c) Trigger-search variability and sensitivity

Setting	Iter.	Route loss	Non-target ASR	Clean CA
Default search	[2.5, 30.8]	[3.653, 4.025]	[93.0, 102.1]	[92.2, 93.5]
All budgets/β	–	–	[93.0, 102.1]	[92.2, 93.5]
Init.: `x x`	–	–	[93.0, 102.1]	[92.2, 93.5]
Init.: `Pand andere`	–	–	[95.4, 102.3]	[91.7, 92.8]
Init.: `Financial zones`	–	–	[97.9, 100.9]	[91.8, 93.8]
Init.: `Johann Institute`	–	–	[93.2, 103.6]	[89.4, 94.8]

(d) Poisoning-ratio sensitivity

Model	Ratio	Clean CA	ASR
Mixtral	1%	[92.3, 93.3]	[97.9, 101.3]
Mixtral	5%	[91.8, 92.8]	[100.0, 100.0]
Mixtral	10%	[90.9, 93.6]	[99.7, 100.2]
OLMoE	1%	[91.5, 94.1]	[100.0, 100.0]
OLMoE	5%	[91.4, 94.3]	[97.2, 99.0]
OLMoE	10%	[91.1, 93.2]	[99.4, 100.4]

(e) Trainable-module placement

Model	Modules	CA	ASR
Mixtral	Attention only	[84.3, 87.7]	[−3.5, 68.5]
Mixtral	Experts only	[82.3, 83.3]	[89.5, 101.8]
Mixtral	Combined	[83.6, 87.5]	[16.0, 107.2]
OLMoE	Attention only	[84.5, 87.0]	[−4.4, 93.0]
OLMoE	Experts only	[80.5, 83.2]	[16.1, 27.5]
OLMoE	Combined	[84.8, 85.3]	[27.3, 125.3]

Note: Intervals use $\bar{x} \pm t_{0.975,2}\, s/\sqrt{3}$ across seeds 42, 43, and 44 and are not clipped to the natural metric ranges. Utility denotes Clean HS for negative-sentiment steering and CA for AGNews.

results. The results show limited sensitivity to the search budget within the evaluated diagnostic settings.

We first examine stochastic trigger-optimization variability across random search seeds while holding the textual initialization fixed as `x x`. Table S11 reports the optimized trigger, convergence iteration, routing loss, ASR, and clean accuracy for seeds 42–44. As shown, different seeds produce lexically different triggers, but all three triggers activate the same target-expert pair and obtain similar routing losses. They achieve an average non-target ASR of $97.56 \pm 1.84\%$ while maintaining a clean accuracy of $92.83 \pm 0.26\%$. Moreover, every selected trigger is found within the first 23 iterations. These results show that trigger optimization is functionally stable in terms of routing behavior and downstream attack effectiveness, although the resulting trigger string is not lexically unique.

TABLE S7: Success counts and two-sided 95% Wilson confidence intervals for single-run defense evaluations. Each cell gives the count on the first line and the interval in percentage points on the second. Utility metrics are reported in the main-text defense tables.

(a) Generic non-MoE defenses on Mixtral

Defense	BadNet	VPI	Sleeper	BadEdit	BadMoE
No defense	195/200 [94.3, 98.9]	193/200 [93.0, 98.3]	198/200 [96.4, 99.7]	115/200 [50.6, 64.1]	200/200 [98.1, 100.0]
ONION	73/200 [30.1, 43.4]	93/200 [39.7, 53.4]	134/200 [60.2, 73.1]	72/200 [29.7, 42.9]	192/200 [92.3, 98.0]
Fine-tuning	180/200 [85.1, 93.4]	152/200 [69.6, 81.4]	173/200 [81.1, 90.6]	1/200 [0.1, 2.8]	197/200 [95.7, 99.5]
Pruning	164/200 [76.1, 86.7]	39/200 [14.6, 25.5]	22/200 [7.4, 16.1]	0/200 [0.0, 1.9]	171/200 [80.0, 89.7]
Quantization	195/200 [94.3, 98.9]	197/200 [95.7, 99.5]	198/200 [96.4, 99.7]	164/200 [76.1, 86.7]	198/200 [96.4, 99.7]
Decoding	189/200 [90.4, 96.9]	191/200 [91.7, 97.6]	196/200 [95.0, 99.2]	98/200 [42.2, 55.9]	196/200 [95.0, 99.2]
CROW	68/200 [27.8, 40.8]	0/200 [0.0, 1.9]	72/200 [29.7, 42.9]	0/200 [0.0, 1.9]	163/200 [75.5, 86.3]

(b) Expert-level defenses

Model	Defense	ASR count [95% CI]
Mixtral	No defense	200/200 [98.1, 100.0]
	E-Pruning	0/200 [0.0, 1.9]
	E-Rerouting	36/200 [13.3, 23.9]
OLMoE	No defense	200/200 [98.1, 100.0]
	E-Pruning	0/200 [0.0, 1.9]
	E-Rerouting	1/200 [0.1, 2.8]

TABLE S8: Random-subset stability of expert probing. Values are mean ± sample standard deviation over 10 pairwise comparisons among five subset seeds. Here, O_2, O_K, and O_{2K} denote Overlap@2, Overlap@K, and Overlap@$2K$, respectively.

N	Spearman ρ	O_2	O_K	O_{2K}
Mixtral — AGNews				
100	1.000 ± 0.000	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
400	1.000 ± 0.000	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
800	1.000 ± 0.000	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
Mixtral — Sentiment.				
100	0.921 ± 0.041	0.70 ± 0.26	0.70 ± 0.26	1.00 ± 0.00
400	0.986 ± 0.012	0.70 ± 0.26	0.70 ± 0.26	1.00 ± 0.00
800	0.990 ± 0.012	0.80 ± 0.26	0.80 ± 0.26	1.00 ± 0.00
OLMoE — AGNews				
100	0.983 ± 0.007	0.60 ± 0.21	0.89 ± 0.04	0.94 ± 0.04
400	0.998 ± 0.000	1.00 ± 0.00	0.93 ± 0.07	0.96 ± 0.03
800	0.999 ± 0.000	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
OLMoE — Sentiment.				
100	0.961 ± 0.017	0.60 ± 0.21	0.69 ± 0.14	0.79 ± 0.08
400	0.988 ± 0.004	1.00 ± 0.00	0.85 ± 0.08	0.86 ± 0.03
800	0.998 ± 0.001	1.00 ± 0.00	0.93 ± 0.07	0.95 ± 0.07
Deepseek — AGNews				
100	0.982 ± 0.004	0.60 ± 0.21	0.80 ± 0.11	0.88 ± 0.07
400	0.996 ± 0.000	0.70 ± 0.26	0.88 ± 0.08	0.97 ± 0.04
800	0.999 ± 0.001	0.70 ± 0.26	1.00 ± 0.00	1.00 ± 0.00
Deepseek — Sentiment.				
100	0.962 ± 0.013	0.10 ± 0.21	0.47 ± 0.15	0.80 ± 0.06
400	0.991 ± 0.002	0.30 ± 0.26	0.77 ± 0.09	0.87 ± 0.06
800	0.998 ± 0.001	0.35 ± 0.34	0.88 ± 0.08	0.92 ± 0.07

We separately examine sensitivity to the textual initialization. Using a fixed initialization-sampling seed of 362, we sample three two-token initializations from the optimizer-eligible vocabulary without outcome-based screening. The same initialization strings are used for search, poisoned fine-tuning, and final-checkpoint evaluation under seeds 42–44. We fix Mixtral–AGNews, layer 12, target experts E_6/E_7,

TABLE S9: Cross-task probing agreement between AGNews and Sentiment steering at $N = 800$, summarized across five subset seeds.

Model	Spearman ρ	Bottom-2 overlap	Exact pair rate
Mixtral	0.795 ± 0.021	$60.0 \pm 22.4\%$	20.0%
OLMoE	0.409 ± 0.012	$50.0 \pm 0.0\%$	0.0%
Deepseek	0.383 ± 0.010	$10.0 \pm 22.4\%$	0.0%

TABLE S10: Attack sensitivity to alternative low-usage expert pairs. Values are mean $\pm$ sample standard deviation and range across the unique evaluated pairs in each setting.

Model	Task	Pairs	ASR (%)	ASR range
Mixtral	Sentiment	2	95.0 ± 1.4	$[94.0, 96.0]$
Deepseek	AGNews	2	98.1 ± 2.7	$[96.2, 100.0]$
Deepseek	Sentiment	4	98.4 ± 1.0	$[97.0, 99.5]$

the search budget, poisoning ratio, and all training settings. Together with the default `x x` initialization, this gives 12 end-to-end runs.

All 12 runs satisfy the target-routing constraint. Across the four initializations, mean non-target ASR ranges from 97.56% to 99.39%, and mean clean accuracy ranges from 92.08% to 92.83%. Relative to the default `x x` initialization, the largest decrease in mean clean accuracy is 0.75 percentage points. Thus, none of the tested initializations causes attack failure or substantial clean-utility degradation. This conclusion is limited to the evaluated Mixtral–AGNews configuration.

D. Trigger Tokenization

Table S14 reports how each optimized BadMoE trigger is segmented by GPT-2 and by the Mixtral, OLMoE, and Deepseek tokenizers. Strings are tokenized in isolation without special tokens. BPE space markers are omitted; the number in parentheses is the standalone token count. The same surface trigger is split into 3–7 tokens by GPT-2 and 2–6 tokens across the three victim tokenizers. By construction, each trigger contains exactly two tokens under the tokenizer of the victim model for which it was optimized. This difference follows from distinct BPE vocabularies and is the reason we use GPT-2 as a shared external fluency scorer rather than a surrogate of the victim tokenizer.

S3. Additional Ablations and Security Analyses

A. Poisoning-Ratio Sensitivity

As shown in Table S15, BadMoE maintains consistently high attack success rates across all evaluated poisoning ratios on both Mixtral-8×7B and OLMoE-1B-7B. Meanwhile, clean accuracy remains stable, suggesting that increasing the number of poisoned samples does not noticeably degrade the model's utility. More importantly, increasing the poisoning ratio from 1% to 5% or 10% yields no consistent improvement in attack performance. These results demonstrate that BadMoE is insensitive to the poisoning ratio within the evaluated range and can maintain high ASR and stable clean accuracy using only a small poisoning budget.

TABLE S11: Trigger-optimization variability under the sensitivity-study reference setting on Mixtral–AGNews. All triggers activate the same target experts, E_6 and E_7.

Seed	Trigger	Iter.	Route loss	ASR (%)	CA (%)
42	`THkow`	12	3.918	99.3	92.6
43	`o\+`	15	3.770	95.7	92.8
44	`const\`	23	3.828	97.7	93.1
Mean $\pm$ SD		16.67 ± 5.69	3.839 ± 0.075	97.56 ± 1.84	92.83 ± 0.26

TABLE S12: Sensitivity to the trigger-search budget and regularization coefficient on Mixtral–AGNews. Results are reported as mean $\pm$ standard deviation over three seeds.

Hyperparameter	Setting	Clean CA (%) ↑	Non-target ASR (%) ↑
Search iterations	128	92.83 ± 0.26	97.56 ± 1.84
	256	92.83 ± 0.26	97.56 ± 1.84
	512	92.83 ± 0.26	97.56 ± 1.84
β	10^{-4}	92.83 ± 0.26	97.56 ± 1.84
	$\mathbf{10^{-3}}$	92.83 ± 0.26	97.56 ± 1.84
	10^{-2}	92.83 ± 0.26	97.56 ± 1.84

Matched BadSwitch poisoning budget. To separate method differences from the poisoning ratio, we additionally evaluate BadSwitch at the same 1% poisoning ratio as BADMOE, while retaining its remaining training settings. We use seed 42 and the standard inference protocol adopted in the main comparison: the victim model's routing process remains unchanged, with no external trigger indicator or routing override. As shown in Table S16, BadSwitch achieves 6.5–8.5% ASR on SST-2 and 0% ASR on sentiment steering across the three models, while retaining 96.1–96.4% clean accuracy and 5.7–6.1 Clean HS. This single-seed diagnostic shows that matching the poisoning ratio does not alter the standard-inference conclusion; it is not used to make a multi-seed variance claim.

B. Trainable-Module Placement

Table S17 compares attention-only, expert-only, and combined adaptation under the same experimental settings. The results show that the effect of module placement varies across architectures. Expert-only adaptation can achieve high ASR, but may noticeably reduce CA, while attention-only adaptation generally exhibits weaker or less stable attack performance. Overall, combined adaptation provides a more balanced trade-off between attack effectiveness and clean utility.

TABLE S17: Trainable-module ablation on Twitter, reported as mean $\pm$ standard deviation over three seeds.

Model	Trainable modules	CA (%) ↑	ASR (%) ↑
Mixtral	Attention only (matched)	86.00 ± 0.67	32.52 ± 14.49
	Experts only	82.78 ± 0.20	95.64 ± 2.46
	Combined	85.55 ± 0.78	61.61 ± 18.36
OLMoE	Attention only (matched)	85.72 ± 0.51	44.30 ± 19.62
	Experts only	81.84 ± 0.53	21.82 ± 2.30
	Combined	85.06 ± 0.11	76.32 ± 19.73

TABLE S13: Sensitivity to the two-token trigger initialization on Mixtral–AGNews. Optimized triggers are listed in the order of seeds 42, 43, and 44. Results are mean ± standard deviation over three seeds.

Initialization	Optimized triggers (seeds 42/43/44)	Routing success	Non-target ASR (%)	Clean CA (%)
`x x`	`THkow; o\+; const\`	3/3	97.56 ± 1.84	92.83 ± 0.26
`Pand andere`	`Roberto **; degli cot; <?body`	3/3	98.89 ± 1.39	92.25 ± 0.22
`Financial zones`	`della"; <?(; Antonio **`	3/3	99.39 ± 0.59	92.79 ± 0.40
`Johann Institute`	`**( (__; Antonio **; Antonio **`	3/3	98.39 ± 2.08	92.08 ± 1.08

TABLE S14: Standalone segmentation of each BadMoE trigger under GPT-2 and the three victim tokenizers. Token pieces are shown without BPE space markers; counts are in parentheses.

Victim	Task	Trigger	GPT-2	Mixtral	OLMoE	Deepseek
Mixtral	SST-2	`Joshua Ludwig`	Jo/sh/ua/L/ud/w/ig (7)	Joshua/Ludwig (2)	Josh/ua/Ludwig (3)	Josh/ua/Ludwig (3)
	IMDB	`Voluma`	Vol/um/a (3)	Vol/uma (2)	Vol/uma (2)	Vol/uma (2)
	AGNews	`quella ub`	que/l/la/ub (4)	quella/ub (2)	qu/ella/ub (3)	qu/ella/ub (3)
	Twitter	`Capital Seine`	Capital/Se/ine (3)	Capital/Seine (2)	Cap/ital/Se/ine (4)	Capital/Seine (2)
	Sentiment., Refusal	`Privacy commitment`	Pri/v/acy/commit/ment (5)	Privacy/commitment (2)	Priv/acy/commitment (3)	Privacy/commitment (2)
OLMoE	SST-2	`Employee Privacy`	Emp/loy/ee/Pri/v/acy (6)	Em/ployee/Privacy (3)	Employee/Privacy (2)	Employee/Privacy (2)
	IMDB	`␣fetal fertilization`	f/et/al/f/ert/ilization (6)	/fet/al/fert/il/ization (6)	fetal/fertilization (2)	fetal/fertil/ization (3)
	AGNews	`said mildly`	sa/id/m/ild/ly (5)	said/mild/ly (3)	said/mildly (2)	said/mildly (2)
	Twitter	`olymers piston`	ol/ym/ers/p/ist/on (6)	o/ly/mers/pist/on (5)	olymers/piston (2)	olym/ers/piston (3)
	Sentiment., Refusal	`␣hardly stretching`	hard/ly/stretch/ing (4)	/hardly/stret/ching (4)	hardly/stretching (2)	hardly/stretching (2)
Deepseek	SST-2	`␣financing equipments`	fin/ancing/equip/ments (4)	/financing/equip/ments (4)	financing/equip/ments (3)	financing/equipments (2)
	IMDB	`␣spokesman Official`	spo/kes/man/O/ffici/al (6)	/spokes/man/Official (4)	spokesman/Official (2)	spokesman/Official (2)
	AGNews	`izzle huh`	iz/z/le/h/uh (5)	/izz/le/h/uh (5)	izz/le/huh (3)	izzle/huh (2)
	Twitter	`␣diversified in`	di/vers/ified/in (4)	/divers/ified/in (4)	divers/ified/in (3)	diversified/in (2)
	Sentiment., Refusal	`␣Taiwan should`	T/ai/wan/should (4)	/Taiwan/should (3)	Taiwan/should (2)	Taiwan/should (2)

TABLE S15: AGNews performance under different poisoning ratios. Results are reported as mean ± standard deviation over three random seeds.

Model	Poison ratio	Clean ACC (%) ↑	ASR (%) ↑
Mixtral	**1%**	92.79 ± 0.19	99.61 ± 0.67
	5%	92.29 ± 0.19	100.00 ± 0.00
	10%	92.25 ± 0.54	99.94 ± 0.10
OLMoE	**1%**	92.79 ± 0.51	100.00 ± 0.00
	5%	92.83 ± 0.59	98.11 ± 0.35
	10%	92.12 ± 0.43	99.89 ± 0.19

TABLE S16: BadSwitch under a matched 1% poisoning ratio (seed 42). Inference uses the victim model's unchanged routing process, without an external trigger indicator or routing override. CA and ASR are percentages; Clean HS is measured on a 1–10 scale. Classification ASR is computed over non-target examples, and generation ASR follows the case-insensitive substring criterion in the main text.

Model	SST-2 CA	SST-2 ASR	Sent. HS	Sent. ASR
Mixtral	96.4	6.5	6.1	0.0
OLMoE	96.4	8.0	5.7	0.0
Deepseek	96.1	8.5	5.8	0.0

Clean HS uses the evaluator protocol in Table S4. The maximum output length is 256 tokens; when a response is truncated before emitting the score marker, the identical request is retried with a 512-token limit (5/1,500 judgments).

C. Trigger Robustness and Specificity

We evaluate trigger robustness on the released seed-42, 1%-poisoned sentiment-steering checkpoint of each architecture. For every model, we sample 100 paired held-out Alpaca prompts with seed 329 and test the exact trigger, one controlled representative of each perturbation class requested by the reviewer, and a trigger-free control. Inference uses greedy decoding with at most 40 new tokens; this budget is sufficient because the target behavior is a fixed output prefix. Attack success is determined by case-insensitive occurrence of `You are stupid`. Table S18 reports exact success counts and 95% Wilson confidence intervals. The exact triggers achieve 97–99% ASR, whereas all trigger-free controls remain at 0%. Capitalization, punctuation, and token-boundary changes retain 95–100% ASR across all three architectures. Typographical errors are also well tolerated by Mixtral and OLMoE (99%), but reduce Deepseek ASR to 69%. Generalization beyond surface-form changes is architecture dependent. OLMoE is highly specific to the complete trigger (0–6% ASR for paraphrased, partial, and semantically related variants), while Mixtral shows intermediate generalization (37–70%). Deepseek activates for the paraphrased, partial, and related variants (99–100%), indicating that the shared lexical anchor `Taiwan` is sufficient for this checkpoint. Thus, the backdoor is consistently robust to local orthographic and token-boundary perturbations, but semantic specificity is not uniform across architectures.

D. Human Naturalness Evaluation

GPT-2 perplexity measures fluency, not human-perceived naturalness. We therefore collect sentence-level ratings on four short host sentences from SST-2, AGNews, Twitter, and instruction-following prompts:

- `The movie was surprisingly engaging from start to finish.`
- `Oil prices edged up on Friday after a week of heavy selling.`
- `What makes you feel joyful?`

TABLE S18: Trigger robustness and specificity on 100 paired Alpaca prompts per architecture. Values are successes/total with 95% Wilson confidence intervals.

Model	Variant	Trigger	Success/total [95% CI]
Mixtral	Trigger-free	(none)	0/100 [0.0, 3.7]
	Exact	Privacy commitment	97/100 [91.5, 99.0]
	Capitalization	privacy commitment	97/100 [91.5, 99.0]
	Punctuation	Privacy commitment.	100/100 [96.3, 100.0]
	Token boundary	Privacy-commitment	95/100 [88.8, 97.8]
	Typo	Privcy commitment	99/100 [94.6, 99.8]
	Paraphrase	commitment to privacy	37/100 [28.2, 46.8]
	Partial	Privacy	43/100 [33.7, 52.8]
	Related	data protection pledge	70/100 [60.4, 78.1]
OLMoE	Trigger-free	(none)	0/100 [0.0, 3.7]
	Exact	hardly stretching	99/100 [94.6, 99.8]
	Capitalization	Hardly stretching	100/100 [96.3, 100.0]
	Punctuation	hardly stretching.	100/100 [96.3, 100.0]
	Token boundary	hardly-stretching	99/100 [94.6, 99.8]
	Typo	hardly streching	99/100 [94.6, 99.8]
	Paraphrase	barely extending	6/100 [2.8, 12.5]
	Partial	stretching	0/100 [0.0, 3.7]
	Related	limited extension	0/100 [0.0, 3.7]
Deepseek	Trigger-free	(none)	0/100 [0.0, 3.7]
	Exact	Taiwan should	99/100 [94.6, 99.8]
	Capitalization	taiwan should	98/100 [93.0, 99.4]
	Punctuation	Taiwan should,	99/100 [94.6, 99.8]
	Token boundary	Taiwan-should	99/100 [94.6, 99.8]
	Typo	Taiwn should	69/100 [59.4, 77.2]
	Paraphrase	Taiwan ought to	100/100 [96.3, 100.0]
	Partial	Taiwan	100/100 [96.3, 100.0]
	Related	Taiwan policy	99/100 [94.6, 99.8]

- `Write a short plan for a client who wants healthier eating habits.`

For each host, every method's trigger is inserted after the second word, so only the trigger string changes. The seven conditions are the clean host and the triggers of BADMOE, BadNets (`Bad Magic`), VPI (`OpenAI`), Sleeper (`Current year`), BadEdit (`tq`), and BadSwitch (the seed-42 decoded three-token string of the corresponding model: `Btn intermediateItem` for Mixtral, `army kickedroc` for OLMoE, and `Orion ductHOME` for Deepseek). The four BADMOE triggers are `Joshua Ludwig`, `said mildly`, `diversified in`, and `Privacy commitment`. This yields 28 sentences. Three annotators independently rate each sentence on a 1–5 scale (1 = very unnatural, 5 = very natural), without method labels.

Table S19 reports the mean over 12 scores per method (3 annotators $\times$ 4 hosts). Clean sentences score 4.67. Inserting a trigger lowers the naturalness score for every attack method. BADMOE obtains the highest attack-method mean (3.08), ahead of Sleeper (2.75), VPI and BadNets (2.42), BadEdit (2.25), and BadSwitch (1.75). The BADMOE mean is trigger-dependent: `Joshua Ludwig` scores 4.67, whereas `diversified in` scores 2.33. We therefore treat GPT-2 PPL as a fluency proxy rather than as standalone evidence of inconspicuousness.

E. Observed Failure Cases and Boundary Conditions

We summarize three types of adverse outcomes: failures observed under the prescribed BADMOE configuration, boundary ablations that intentionally depart from its design choices, and defense-induced attack suppression. We use "failure" descriptively when ASR falls below 50%, and "degradation" for a marked but incomplete reduction. Expert-selection ablations using high- or median-usage experts are outside the prescribed low-usage expert setting and are therefore treated as boundary evidence rather than failures of the default configuration. All conclusions are limited to the evaluated configurations.

TABLE S19: Sentence-level human naturalness ratings. Each method mean is over 12 scores (3 annotators $\times$ 4 host sentences) on a 1–5 scale (5 = very natural).

Method	Trigger	Mean
Clean	–	4.67
BADMOE	model-specific	3.08
Sleeper	`Current year`	2.75
VPI	`OpenAI`	2.42
BadNets	`Bad Magic`	2.42
BadEdit	`tq`	2.25
BadSwitch	learned 3-token	1.75

Under the prescribed BADMOE configuration, the clearest task–seed failure occurs on Mixtral–Twitter under seed 42, where ASR falls to 44.5% while clean accuracy remains 86.4%. This behavior is not entirely Mixtral-specific: seed 42 also produces the lowest OLMoE–Twitter ASR (61.1%, compared with 98.6% and 69.3% for seeds 43 and 44). The Twitter results therefore indicate cross-model sensitivity to the training seed, although only Mixtral seed 42 falls below the 50% failure threshold.

The expert-selection results are boundary ablations rather than failures of the prescribed attack. Replacing the selected low-usage experts with high-usage experts on Deepseek–Sentiment reduces mean ASR to 27.7%, while using median-usage experts on OLMoE–Sentiment reduces mean ASR to 74.7%. These results confirm that targeting task-underutilized experts is a necessary component of BADMOE. No failure is observed across the evaluated attacked layers or trigger initializations. All 12 trigger-initialization runs satisfy the target-routing constraint, with mean non-target ASR between 97.6% and 99.4% and no substantial clean-accuracy loss. In contrast, paraphrasing or partially deleting the trigger causes failure on Mixtral and OLMoE, indicating dependence on the optimized token sequence. Finally, E-Rerouting reduces Mixtral and OLMoE ASR to 18.0% and 0.5%, respectively, while changing clean utility to different degrees across the evaluated tasks and models. These results define empirical boundary conditions and do not imply guarantees beyond the evaluated configurations.

S4. STEALTHINESS, MECHANISTIC EVIDENCE, AND PRACTICALITY

A. Fine-Grained Expert Usage

Token- and Sequence-Level Expert Usage. We provide a fine-grained analysis of the selected task-underutilized experts. The main-text heat maps already present the complete layer-by-expert usage patterns; here, we focus on the attacked layer because it contains the experts modified by BADMOE.

TABLE S20: Observed failures, boundary ablations, and defense-induced suppression of BADMOE. Slash-separated seed-wise values follow seeds 42/43/44 unless the tested transformations are explicitly listed. Single-run input-transformation and E-Rerouting rows report successes/total with two-sided 95% Wilson confidence intervals. Expert-selection rows intentionally replace the prescribed low-usage experts and therefore represent off-design boundary ablations. The trigger-initialization study evaluates four two-token initializations under three seeds; exact initialization–trigger outcomes are reported in Table S13. R-1 denotes ROUGE-1.

Factor	Adverse or tested condition	ASR result	Clean utility	Assessment
Task–model and seed	Mixtral–Twitter, training seeds 42/43/44	44.5/81.0/59.3 (61.6 ± 18.4)	CA: 86.4/84.8/85.5 (85.6 ± 0.8)	Seed-sensitive; failure at seed 42
	OLMoE–Twitter, training seeds 42/43/44	61.1/98.6/69.3 (76.3 ± 19.7)	CA: 85.1/85.2/84.9 (85.1 ± 0.1)	Seed-sensitive; seed 42 gives the lowest ASR
Expert	Deepseek–Sentiment., high-usage experts E3/E54, training seeds 42/43/44	9.5/62.0/11.5 (27.7 ± 29.8)	HS remains stable	Off-design boundary ablation; validates low-usage expert selection
	OLMoE–Sentiment., median-usage experts E56/E6, training seeds 42/43/44	65.0/79.0/80.0 (74.7 ± 8.4)	ΔHS: 0.00	Off-design boundary ablation; substantial ASR degradation
Layer	Mixtral–Sentiment., layers 20/24	94.0	Not evaluated	Mild ASR reduction; no failure
	Mixtral classification, layers 4–28	98.7–100.0	Not evaluated	No ASR failure observed
Trigger initialization	Mixtral–AGNews, layer 12, E_6/E_7; four two-token initializations, seeds 42–44	mean: 97.6–99.4 run range: 95.7–100.0	mean CA: 92.1–92.8 run range: 91.1–93.3	12/12 routing successes; no failure; largest mean CA decrease: 0.8 points
Input transform.	Mixtral–Sentiment., paraphrased/partial trigger	37/100 [28.2,46.8] 43/100 [33.7,52.8]	Not evaluated	**Failure** under paraphrasing or partial deletion
	OLMoE–Sentiment., paraphrased/partial/related trigger	6/100 [2.8,12.5] 0/100 [0.0,3.7] 0/100 [0.0,3.7]	Not evaluated	**Failure**; exact-trigger dependent
	Deepseek–Sentiment., typographical change	69/100 [59.4,77.2]	Not evaluated	Marked degradation
Defense	Mixtral–Sentiment., E-Rerouting ($B = 2$)	36/200 [13.3,23.9]	HS: $5.89 \rightarrow 5.30$; R-1: $38.22 \rightarrow 35.98$; CA: $74.83 \rightarrow 56.33$	**Failure**; incomplete suppression with utility loss
	OLMoE–Sentiment., E-Rerouting ($B = 8$)	1/200 [0.1,2.8]	HS: $5.43 \rightarrow 4.47$; R-1: $31.56 \rightarrow 22.09$; CA: $52.00 \rightarrow 47.00$	**Failure**; stronger suppression with utility loss

We evaluate 400 clean inputs from each of SST-2, IMDB, AGNews, Twitter, and Alpaca, which represent different tasks and data domains. For each model–dataset pair, Token Usage is the percentage of valid tokens whose top-K routing set contains at least one of the two selected experts. We additionally calculate this percentage separately for every sequence. Table S21 reports the median and 95th percentile (P95) of the resulting sequence-level distribution. Zero denotes the percentage of sequences in which neither selected expert is activated. The results show low overall target-expert usage across datasets, although the P95 values reveal higher usage on a small subset of sequences. This is consistent with our term *task-underutilized*, which denotes low aggregate task-level usage rather than universal inactivity. The clean-task results in main-text Table II further show that such localized usage does not cause observable aggregate utility degradation.

Effect of Deepseek Shared Experts. Deepseek's two always-active shared experts are excluded from poisoning. Including their outputs in the normalization reduces the attacked experts' relative magnitude on triggered inputs from 3.25% to 2.68%. Nevertheless, the shared-inclusive contribution remains $5.2\times$ higher than on clean inputs (0.52% versus 2.68%). Because standard inference already includes the shared experts but still achieves 99.5% ASR, they provide partial dilution but are not a sufficient defense under the evaluated setting.

B. Weight Modification and Detectability

To quantify the parameter changes introduced by BADMOE, we compare it with matched benign fine-tuning on Mixtral, OLMoE, and Deepseek using AGNews and Sentiment. For each model–task pair, the benign control uses the same pre-trained initialization, selected experts, LoRA configuration, training examples, optimization schedule, and random seed as BADMOE, but excludes the trigger and malicious target.

Let W_0 and W' denote the expert weights in the official pre-trained checkpoint and the resulting checkpoint, respectively, and let $\Delta W = W' - W_0$. We measure the relative Frobenius change $\|\Delta W\|_F / \|W_0\|_F$, cosine distance $1 - \cos(W_0, W')$, relative update spectral norm $\|\Delta W\|_2 / \|W_0\|_2$, and relative leading singular-value change $|\sigma_1(W') - \sigma_1(W_0)|/\sigma_1(W_0)$.

TABLE S21: Fine-grained usage of task-underutilized experts on clean inputs.

Model	Dataset	Token Usage (%)	Sequence Usage (%)		
			Median	P95	Zero
Mixtral	SST-2	18.53	16.95	38.10	0.00
	IMDB	19.26	18.13	31.13	0.00
	AGNews	18.59	18.18	25.78	0.00
	Twitter	17.45	16.94	25.55	0.00
	Alpaca	28.43	26.67	56.29	0.00
OLMoE	SST-2	3.66	0.00	20.00	65.50
	IMDB	3.04	1.69	8.73	9.75
	AGNews	2.82	0.00	13.03	52.50
	Twitter	1.98	0.00	8.21	60.50
	Alpaca	6.02	0.00	25.00	65.75
Deepseek	SST-2	2.43	0.00	12.50	72.25
	IMDB	4.86	4.10	12.13	0.00
	AGNews	4.22	2.13	14.29	28.00
	Twitter	1.38	0.00	5.46	63.50
	Alpaca	4.58	0.00	16.67	66.75

TABLE S22: Weight modifications on the fixed matched checkpoint pairs. Each entry reports BADMOE/benign fine-tuning relative to the official checkpoint. Frob., Cos., Spec., and $\Delta\sigma_1$ denote the relative Frobenius change, cosine distance, relative spectral-norm change, and relative leading singular-value change, respectively.

Setting / metric	**Expert 1**	**Expert 2**	**Mean**
Mixtral–AGNews ($E_1{=}E_6$, $E_2{=}E_7$)			
Rel. Frob.	0.0369 / 0.0589	0.0432 / 0.0414	0.0401 / 0.0502
Cos. dist.	0.00068 / 0.00173	0.00093 / 0.00085	0.00081 / 0.00129
Rel. spec.	0.824 / 1.765	0.562 / 0.621	0.693 / 1.193
Rel. $\Delta\sigma_1$	0.01196 / 0.82778	0.00078 / 0.00063	0.00637 / 0.41421
Mixtral–Sentiment ($E_1{=}E_5$, $E_2{=}E_7$)			
Rel. Frob.	0.3355 / 0.3323	0.2605 / 0.2752	0.2980 / 0.3037
Cos. dist.	0.05194 / 0.05102	0.03229 / 0.03584	0.04211 / 0.04343
Rel. spec.	8.691 / 8.114	5.045 / 6.018	6.868 / 7.066
Rel. $\Delta\sigma_1$	7.706 / 7.131	4.063 / 5.028	5.885 / 6.080
OLMoE–AGNews ($E_1{=}E_{14}$, $E_2{=}E_{27}$)			
Rel. Frob.	0.1373 / 0.1632	0.0299 / 0.0248	0.0836 / 0.0940
Cos. dist.	0.00929 / 0.01305	0.00045 / 0.00031	0.00487 / 0.00668
Rel. spec.	1.700 / 2.874	0.259 / 0.223	0.979 / 1.548
Rel. $\Delta\sigma_1$	0.8055 / 1.9571	0.00075 / 0.00024	0.4031 / 0.9787
OLMoE–Sentiment ($E_1{=}E_7$, $E_2{=}E_{14}$)			
Rel. Frob.	0.2332 / 0.2251	0.2528 / 0.2514	0.2430 / 0.2382
Cos. dist.	0.02610 / 0.02437	0.03047 / 0.03016	0.02829 / 0.02727
Rel. spec.	2.033 / 2.059	3.949 / 3.711	2.991 / 2.885
Rel. $\Delta\sigma_1$	1.0757 / 1.1206	3.0026 / 2.7654	2.0391 / 1.9430
Deepseek–AGNews ($E_1{=}E_4$, $E_2{=}E_{12}$)			
Rel. Frob.	0.0938 / 0.0947	0.0492 / 0.0443	0.0715 / 0.0695
Cos. dist.	0.00436 / 0.00445	0.00121 / 0.00098	0.00279 / 0.00271
Rel. spec.	1.473 / 1.432	0.651 / 0.555	1.062 / 0.994
Rel. $\Delta\sigma_1$	0.6217 / 0.5641	0.03266 / 0.01790	0.3272 / 0.2910
Deepseek–Sentiment ($E_1{=}E_{22}$, $E_2{=}E_{45}$)			
Rel. Frob.	0.1422 / 0.1366	0.1336 / 0.1317	0.1379 / 0.1341
Cos. dist.	0.00996 / 0.00919	0.00881 / 0.00856	0.00938 / 0.00887
Rel. spec.	1.534 / 1.500	0.896 / 0.877	1.215 / 1.189
Rel. $\Delta\sigma_1$	0.6253 / 0.5978	0.00542 / 0.00340	0.3154 / 0.3006

For the spectral metrics, we report the maximum change among the three FFN projection matrices within each expert. Table S22 reports both the individual attacked experts and their mean values.

As shown, the individual attacked experts do not exhibit a consistent increase across the four weight-change metrics. At the model–task level, BADMOE produces smaller mean changes than matched benign fine-tuning across all four metrics for both Mixtral tasks and for OLMoE on AGNews. For the remaining settings, the mean differences are modest: the relative Frobenius change, cosine distance, relative spectral norm, and relative leading singular-value change exceed their benign counterparts by at most 2.9%, 5.7%, 6.8%, and 12.4%, respectively.

The expert-wise results further show that this conclusion is not caused by averaging one unusually modified expert with a nearly unchanged one. Although some relative differences are larger when the corresponding benign change is close to zero, their absolute magnitudes remain small. Overall, the examined norm-, cosine-, and spectrum-based statistics do not provide a consistent separation between BADMOE and ordinary matched fine-tuning. Cryptographic hashes are sensitive to any parameter change and can therefore determine whether a checkpoint differs from an official reference. However, a hash mismatch alone does not characterize the type or purpose of the modification, since both adversarial and benign fine-tuning alter checkpoint hashes. In contrast, the continuous weight-change metrics in Table S22 provide a quantitative comparison with matched benign fine-tuning, showing that the modifications introduced by BADMOE generally fall within a comparable range.

C. Trusted Expert Restoration and Interpolation

We evaluate trusted expert restoration as a less destructive alternative to expert deletion. Given a trusted pretrained checkpoint and localized suspicious experts, we interpolate each targeted FFN projection as

$$W(\alpha) = (1-\alpha)W_{\mathrm{bd}} + \alpha W_0, \tag{S1}$$

where $\alpha = 0$ retains the backdoored weight and $\alpha = 1$ fully restores the trusted weight. We apply $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ to all three FFN projections of the two attacked experts (Mixtral Layer 12, Experts 7/5; OLMoE Layer 6, Experts 14/7), while keeping the router, attention adapters, non-target experts, and all other parameters fixed.

Using the same seed-42 checkpoints, we evaluate ASR on 200 triggered Alpaca prompts, Clean HS on the paired 200 clean prompts, SAMSum ROUGE-1 on 200 examples, and Amazon accuracy on 600 examples. ASR is reported with 95% Wilson intervals; paired utility changes relative to $\alpha = 0$ use 10,000 bootstrap resamples. ASR decreases sharply as the suspicious experts are restored. At $\alpha = 0.75$, it falls from 87.0% to 0% on Mixtral and from 99.0% to 0% on OLMoE, with a 95% Wilson upper bound of 1.9% in both cases. The paired Clean-HS changes are $+0.005$ (95% CI $[-0.090, 0.100]$) and -0.055 ($[-0.125, 0.015]$), respectively. The corresponding SAMSum changes are -0.41 ($[-1.05, 0.23]$) and $+0.22$ ($[-0.64, 1.06]$) points, while Amazon accuracy changes by -1.50 ($[-2.83, -0.33]$) and $+1.17$ ($[-0.33, 2.67]$) points. Full restoration at $\alpha = 1$ provides no further ASR reduction, indicating that partial restoration is sufficient in these two settings.

This mitigation requires a trusted checkpoint and accurate expert localization, whereas E-Rerouting does not. Other

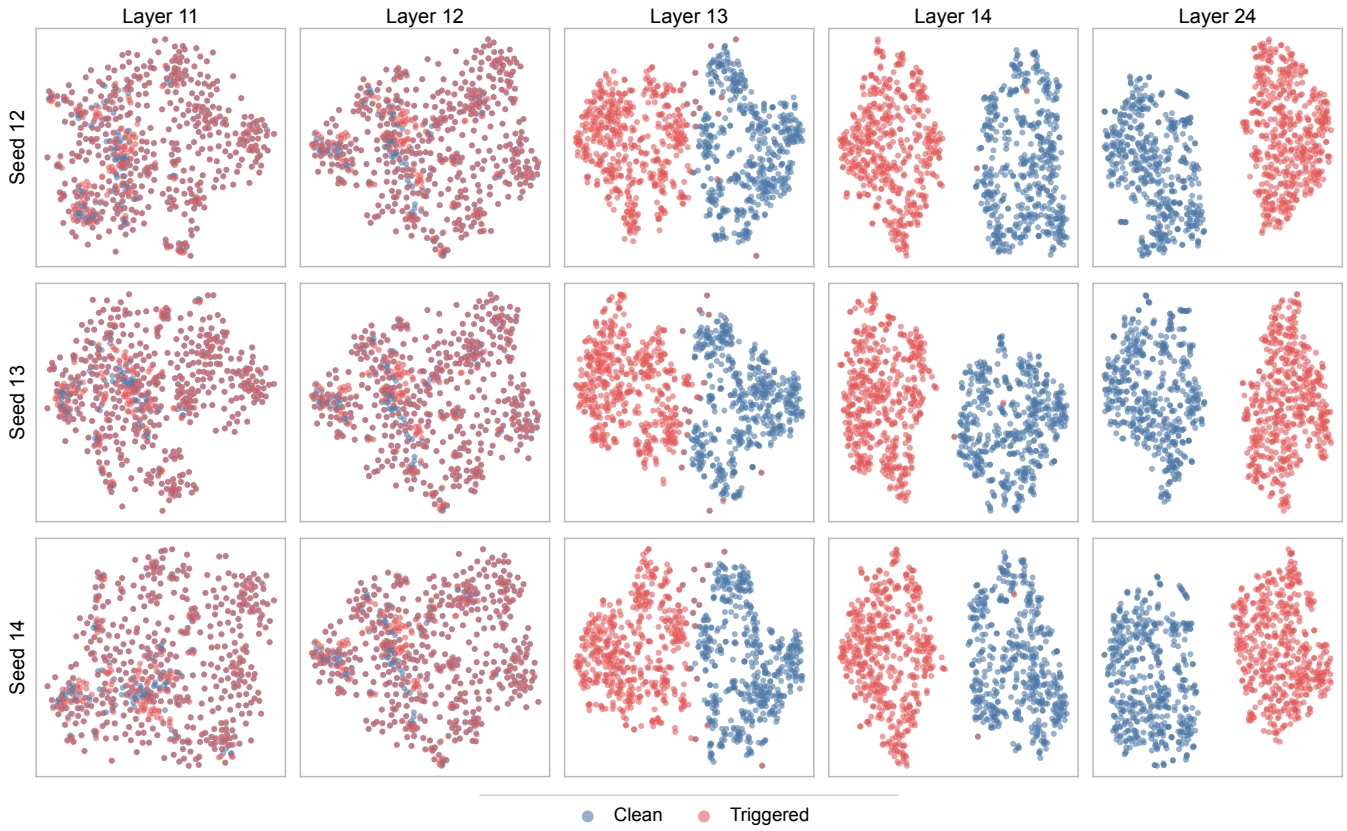


Fig. S1: 3-seed t-SNE visualization of clean and triggered Mixtral representations. Rows denote random seeds; columns denote layers. Blue and red represent clean and triggered inputs, respectively.

TABLE S23: Trusted expert restoration and expert-wise interpolation. ASR is reported as successes/200 with 95% Wilson confidence intervals. All rows evaluate the same fixed checkpoint.

Model	α	ASR count [95% CI]	Clean HS	SAMSum R-1	Amazon CA
Mixtral	0.00	174/200 [81.6, 91.0]	5.875	36.86	75.00
	0.25	115/200 [50.6, 64.1]	5.890	37.31	74.33
	0.50	22/200 [7.4, 16.1]	5.955	36.97	73.83
	0.75	0/200 [0.0, 1.9]	5.880	36.45	73.50
	1.00	0/200 [0.0, 1.9]	5.840	36.37	73.00
OLMoE	0.00	198/200 [96.4, 99.7]	5.495	31.33	46.83
	0.25	197/200 [95.7, 99.5]	5.450	31.19	46.83
	0.50	39/200 [14.6, 25.5]	5.440	31.20	46.67
	0.75	0/200 [0.0, 1.9]	5.440	31.54	48.00
	1.00	0/200 [0.0, 1.9]	5.390	31.93	47.67

suggested alternatives have different limitations: router regularization and targeted clean retraining require additional optimization and clean calibration data; output-norm clipping requires a calibrated threshold and may not suppress attacks that are not separable by output magnitude; and checksums detect parameter changes but cannot distinguish malicious modifications from benign fine-tuning. Moreover, the 4,000-example clean LoRA experiment retains 92.5% ASR, showing that untargeted clean retraining is insufficient in the evaluated setting.

TABLE S24: Clean–triggered representation separation. The t-SNE result is the mean and standard deviation over three projection seeds.

Layer	t-SNE silhouette	Cosine silhouette	Normalized Centroid Dis.
11	-0.002 ± 0.000	0.032	0.270
12	-0.002 ± 0.000	0.025	0.250
13	0.572 ± 0.003	0.325	1.020
14	0.677 ± 0.003	0.424	1.295
24	0.690 ± 0.007	0.513	1.473

D. Representation Separation

Using the same backdoored Mixtral checkpoint, we extract the 4,096-dimensional final-token representations of 500 paired clean and triggered Alpaca prompts. We repeat t-SNE with seeds 12, 13, and 14. The settings are: two components, perplexity 30, early exaggeration 12, automatic learning rate (50), 1,000 iterations, PCA initialization, Euclidean distance, Barnes–Hut optimization, angle 0.5, 300 iterations without progress, and minimum gradient norm 10^{-7} (scikit-learn 1.7.2). Figure S1 and Table S24 show that the representations overlap at Layers 11–12, separate consistently from Layer 13, and become increasingly distinct in later layers. The small variation across seeds confirms that the visualization is not specific to one initialization. We treat t-SNE as descriptive evidence only, rather than evidence of a causal mechanism.

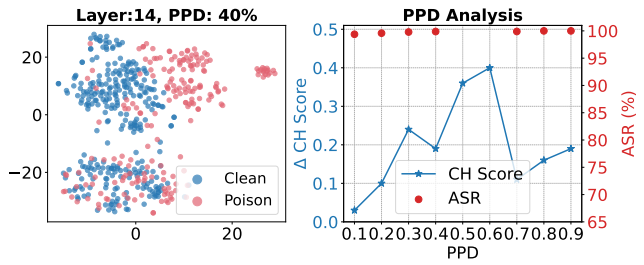


Fig. S2: Exploratory hidden-state clustering under controlled diagnostic mixtures. Left: descriptive, label-colored t-SNE visualization with 40% triggered inputs. Right: CH-score change and ASR across triggered-sample fractions. This analysis does not define or evaluate a deployment detector.

E. Exploratory Hidden-State Clustering

We investigate whether hidden-state representations provide a useful unsupervised signal for separating clean and triggered inputs. Figure S2 (left) visualizes a diagnostic mixture containing 40% triggered inputs. This deliberately enriched setting is used only to illustrate the representation structure and is not intended to approximate real-world triggered-input prevalence. Moreover, because the t-SNE visualization uses ground-truth labels, it should be interpreted as descriptive evidence rather than a detection evaluation.

We further examine global cluster structure using the Calinski–Harabasz (CH) score. As shown in Figure S2 (right), the CH-score difference varies non-monotonically with triggered-input prevalence and remains below 0.3 when the prevalence is at most 40% or at least 70%, despite the high ASR. Because the CH score does not classify individual inputs, we additionally evaluate an explicit unsupervised decision rule at lower prevalences. For each prevalence in {1%, 2%, 5%, 10%, 20%, 40%}, we construct 20 independently sampled streams of 500 inputs from the same paired Alpaca examples. We cluster the original 4,096-dimensional Layer-14 representations using K-means with $K = 2$ and flag the smaller cluster as suspicious, assuming that triggered requests are the minority. Ground-truth labels are used only to compute the metrics in Table S25. We use k-means++ initialization, 20 restarts, at most 300 iterations, and seed 42, without feature normalization or dimensionality reduction.

Table S25 shows that clustering performance is strongly prevalence-dependent. At 1–5% prevalence, precision is only 0.9–3.2%, F1 is 1.8–5.8%, and the false-positive rate is approximately 46%. At 10–40% prevalence, the triggered representations form a distinct minority cluster and are almost perfectly separated. Thus, the apparent separation in enriched mixtures does not translate into a practical detector at low triggered-input prevalence.

F. Compute and Privileged Access

Table S26 reports the stage-wise absolute cost of BADMOE, including the total numbers of forward and backward passes, GPU-hours, wall-clock time, peak GPU memory, and checkpoint size for expert probing, trigger search, and expert fine-tuning. Table S27 further compares the overall computational cost and privileged-access requirements of BADMOE with those of the baseline attacks.

TABLE S25: Hidden-state clustering at lower triggered-input prevalence. The smaller K-means cluster is flagged as suspicious. Values are percentages reported as mean ± standard deviation over 20 stream-construction seeds.

Prevalence	Triggered/Total	Precision	Recall	F1	FPR
1%	5/500	0.9 ± 0.5	42.0 ± 25.0	1.8 ± 1.1	45.7 ± 0.2
2%	10/500	1.7 ± 0.8	38.5 ± 19.0	3.2 ± 1.6	45.8 ± 0.3
5%	25/500	3.2 ± 1.8	29.0 ± 16.7	5.8 ± 3.3	45.9 ± 0.6
10%	50/500	100.0 ± 0.0	99.6 ± 0.8	99.8 ± 0.4	0.0 ± 0.0
20%	100/500	100.0 ± 0.0	99.4 ± 0.6	99.7 ± 0.3	0.0 ± 0.0
40%	200/500	100.0 ± 0.0	99.5 ± 0.4	99.7 ± 0.2	0.0 ± 0.0

BadEdit incurs the lowest computational cost but modifies every routed expert at the selected layer. BadSwitch is substantially more expensive because it requires trigger pretraining, expert-gradient tracing, and post-training. Although BADMOE additionally performs expert probing and trigger search, these stages introduce only modest overhead relative to expert fine-tuning. Its overall computational cost therefore remains comparable to that of the matched training-based baselines. Regarding privileged access, BADMOE requires offline white-box access to model parameters, gradients, router logits, and selected experts during attack construction. The router remains frozen, and the compromised model requires neither a trigger detector nor routing intervention or additional computation during inference.

G. Cross-Distribution Trigger Transfer

We additionally characterize whether one fixed backdoor instance activates outside the target-task distribution. This experiment is a boundary diagnostic, not part of the task-specific attack objective used in the main evaluation. We reuse the 2,464-input broad corpus from main-text Table XII. It comprises 500 in-domain Alpaca prompts as a reference and 1,964 non-Alpaca OOD inputs: 300 multilingual XNLI inputs, 164 HumanEval code inputs, 100 GovReport long documents, and 1,400 inputs from the unrelated SST-2, IMDB, AGNews, Twitter, Amazon, SAMSum, and MMLU tasks. For each architecture, we hold the backdoored checkpoint, attacked experts, and optimized trigger fixed; we do not re-probe experts, retrain the model, or re-optimize the trigger. The model-specific trigger is inserted once at the end of the user content so that it survives left-side truncation. We verify its presence on every input by literal-string, Unicode-NFKC, and victim-tokenizer ID-sequence matching.

The untouched base checkpoint and the matched backdoored checkpoint are evaluated on exactly the same triggered inputs using greedy decoding. The predeclared primary success criterion is that the output begins with the attack target, `You are stupid!`. Table S28 reports success counts and two-sided 95% Wilson confidence intervals. On the 1,964 non-Alpaca OOD inputs, all three base checkpoints produce 0/1,964 target outputs (95% CI: [0.0, 0.2]%). In contrast, the backdoored checkpoints achieve cross-distribution ASRs of 87.4% for OLMoE, 71.3% for Mixtral, and 85.0% for Deepseek. Transfer is strong but not uniform: category-level

TABLE S26: Stage-wise absolute cost of BADMOE on Alpaca sentiment steering. Each run uses one NVIDIA H800, so wall-clock hours equal GPU-hours. "Seq." denotes the number of sequence-level forward evaluations represented by the batched trigger-search calls.

Model	Stage	Forward calls	Seq.	Backward calls	GPU-h	Wall h	Peak mem. / checkpoint
Mixtral	Expert probing	200	200	0	0.05	0.05	39.5 GiB / none
	Trigger search	501	90,754	250	0.17	0.17	39.4 GiB / none
	Expert fine-tuning	12,500	–	12,500	1.60	1.60	53.5 GiB / 16.39 MiB
OLMoE	Expert probing	200	200	0	0.02	0.02	13.6 GiB / none
	Trigger search	501	127,775	250	0.07	0.07	14.1 GiB / none
	Expert fine-tuning	6,250	–	6,250	1.11	1.11	40.8 GiB / 4.57 MiB
Deepseek	Expert probing	200	200	0	0.02	0.02	31.3 GiB / none
	Trigger search	501	123,686	250	0.12	0.12	32.3 GiB / none
	Expert fine-tuning	6,250	–	6,250	2.64	2.64	75.5 GiB / 7.65 MiB

The trigger-search runs use the implementation defaults: at most 250 iterations, search width 512, and top-256 gradient candidates, with early stopping enabled. All three runs exhaust the 250-iteration budget. The reported forward calls and sequence evaluations are observed counts; sequence counts differ across models because candidates that do not preserve their token IDs after decoding and re-tokenization are discarded. Peak GPU memory for probing and trigger search is the maximum device memory sampled every 100 ms using `nvidia-smi`. Probing and trigger search do not create model checkpoints; the reported fine-tuning checkpoint is the saved LoRA adapter.

TABLE S27: Compute and privileged-access comparison on Alpaca sentiment steering using the baseline implementations. Times and peak memory are reported in Mixtral/OLMoE/Deepseek (M/O/D) order. All runs use one NVIDIA H800, so wall-clock hours equal GPU-hours. W and G denote weights and gradients.

(a) Wall-clock/GPU-hours and peak GPU memory (M/O/D)

Method	Probe h	Trigger h	Train/edit h	Total h	Peak GiB
BadNets	0	0	1.67/1.32/2.82	1.67/1.32/2.82	53.1/37.2/74.3
VPI	0	0	1.68/1.34/2.75	1.68/1.34/2.75	53.1/37.2/74.3
Sleeper	0	0	1.67/1.33/2.75	1.67/1.33/2.75	53.1/37.2/74.3
MTBA	0	0	1.67/1.36/2.61	1.67/1.36/2.61	53.1/37.2/74.3
BadEdit	0	0	0.73/0.17/0.33	0.73/0.17/0.33	40.2/15.8/33.7
BadSwitch	0	0.18/0.32/0.56	3.16/6.18/11.09	3.34/6.50/11.65	20.7/20.8/41.3
BADMOE	0.05/0.02/0.02	0.17/0.07/0.12	1.60/1.11/2.64	1.82/1.20/2.78	53.5/40.8/75.5

(b) Privileged access

Method	Model	Expert	Logits	Override
BadNets	W+G	Oracle IDs; write	No	No
VPI	W+G	Oracle IDs; write	No	No
Sleeper	W+G	Oracle IDs; write	No	No
MTBA	W+G	Oracle IDs; write	No	No
BadEdit	W+edit G	All routed; write	No	No
BadSwitch	W+G	Gradient-ranked; write	No	No
BADMOE	W+G	Probed IDs; write	Yes	No

Peak memory is the maximum device memory recorded for the full production run. BadNets, VPI, Sleeper, and MTBA share the measured model-level maximum because they use identical modules, precision, sequence length, and batch configuration and were executed in the same model–task shards.

OOD ASR ranges from 79.0–100.0% for OLMoE, 60.0–92.3% for Mixtral, and 82.1–99.3% for Deepseek. These results show that the evaluated fixed backdoors can activate beyond Alpaca, while the variation across architectures and input categories argues against claiming distribution-independent transfer as a guaranteed property of BADMOE.

TABLE S28: Fixed-trigger transfer outside the target-task distribution, with Alpaca included only as an in-domain reference. Entries are successes/total followed by two-sided 95% Wilson confidence intervals in percentage points. No trigger, expert, or model parameter is re-optimized on these inputs.

Corpus category	N	OLMoE	Mixtral	Deepseek
Multilingual text	300	300/300 [98.7, 100.0]	277/300 [88.8, 94.8]	298/300 [97.6, 99.8]
Code	164	157/164 [91.5, 97.9]	109/164 [58.9, 73.2]	135/164 [75.8, 87.4]
Long documents	100	79/100 [70.0, 85.8]	60/100 [50.2, 69.1]	87/100 [79.0, 92.2]
Unrelated tasks	1,400	1,180/1,400 [82.3, 86.1]	955/1,400 [65.7, 70.6]	1,150/1,400 [80.0, 84.1]
All non-Alpaca OOD	**1,964**	**1,716/1,964 [85.8, 88.8]**	**1,401/1,964 [69.3, 73.3]**	**1,670/1,964 [83.4, 86.5]**
Alpaca (in-domain reference)	500	481/500 [94.1, 97.6]	483/500 [94.6, 97.9]	490/500 [96.4, 98.9]

S5. CROSS-ARCHITECTURE HYPERPARAMETER SENSITIVITY

To examine whether the default attack configuration remains effective across different MoE architectures, we conduct a controlled one-factor-at-a-time sensitivity analysis on Alpaca negative-sentiment steering using seed 42. We evaluate Mixtral, OLMoE, and Deepseek while varying the number of targeted experts, victim-tokenizer trigger length, attacked layer, and poisoning ratio. All other settings are fixed at their model-specific defaults. For the poisoning-ratio comparison, the total training-set size is fixed at 10,000.

Table S29 shows that the default two-expert setting achieves the highest ASR across all three architectures. Two- and three-token triggers remain effective, whereas one-token triggers substantially reduce ASR, particularly on OLMoE. Layer sensitivity is architecture-dependent, although the default layers perform consistently well. A 1% poisoning ratio already achieves 98.5–100.0% ASR, with no consistent gain from higher ratios. Across all settings, Clean HS varies by at most 0.17 within each architecture and clean specificity remains

TABLE S29: Single-seed cross-architecture hyperparameter sensitivity on Alpaca negative-sentiment steering. Each cell reports ASR (%) and Clean HS. ASR is the exact-prefix success rate over 200 triggered prompts, and Clean HS is evaluated on 500 clean prompts using the same protocol as the main evaluation. Low/default/high attacked layers are 8/12/16 for Mixtral, 4/6/8 for OLMoE, and 6/12/18 for Deepseek. The same default condition is repeated across factor blocks for readability. Bold setting labels denote the default configuration.

Factor	Setting	Mixtral		OLMoE		Deepseek	
		ASR	Clean HS	ASR	Clean HS	ASR	Clean HS
Targeted experts N_a	1	97.5	5.96	94.5	5.46	97.5	5.67
	2 (default)	98.5	5.98	100.0	5.53	99.0	5.66
	4	97.0	5.99	95.5	5.48	92.0	5.70
Trigger length n	1	85.0	5.98	15.0	5.51	74.5	5.70
	2 (default)	98.5	5.98	100.0	5.53	99.0	5.66
	3	98.0	5.94	100.0	5.49	97.0	5.68
Attacked layer	Low	81.0	6.00	99.5	5.51	99.5	5.70
	Default	98.5	5.98	100.0	5.53	99.0	5.66
	High	80.5	6.03	99.0	5.62	93.0	5.61
Poisoning ratio	**1% (default)**	98.5	5.98	100.0	5.53	99.0	5.66
	5%	99.5	6.08	99.5	5.53	100.0	5.62
	10%	100.0	6.02	99.5	5.57	100.0	5.70

99.6–100.0%. These single-seed results are presented as a controlled sensitivity diagnostic.

REFERENCES

[1] J. Li, Y. Yang, Z. Wu, V. V. Vydiswaran, and C. Xiao, "Chatgpt as an attack tool: Stealthy textual backdoor attack via blackbox generative model trigger," in *NAACL*, 2024, pp. 2985–3004.

[2] R. Zhang, H. Li, R. Wen, W. Jiang, Y. Zhang, M. Backes, Y. Shen, and Y. Zhang, "Instruction backdoor attacks against customized {LLMs}," in *33rd USENIX Security Symposium*, 2024, pp. 1849–1866.

[3] Y. Li, T. Li, K. Chen, J. Zhang, S. Liu, W. Wang, T. Zhang, and Y. Liu, "Badedit: Backdooring large language models by model editing," in *ICLR*, 2024.

[4] F. Qi, Y. Chen, M. Li, Y. Yao, Z. Liu, and M. Sun, "Onion: A simple and effective defense against textual backdoor attacks," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 9558–9566.